



**BITS Pilani**  
**DIGITAL**

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 3

# Types of Financial Securities



STUDY NOTES • EDITION 1.0 • WEEK 3

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# How to use these notes



These notes cover the whole of Week 3: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A covers the two traditional securities — stocks and bonds — and compares the markets they trade in. Part B introduces derivatives, whose value is borrowed from something else entirely. Part C turns to the pooled vehicles through which most people actually hold all of the above. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

## What each topic contains

Every topic follows the same rhythm, so you always know where to look.

| Section          | What it gives you                                                     |
|------------------|-----------------------------------------------------------------------|
| In one line      | The whole idea compressed to a single sentence                        |
| Why this matters | Where the concept sits in the course, and why it earns your attention |
| The core idea    | The full explanation, developed step by step                          |
| Figure           | A diagram or comparison that makes the idea visible                   |
| Callouts         | Key ideas, worked examples, pitfalls and exam pointers                |
| Quick recap      | Five bullets to revise from                                           |

## Callouts used in this document

### KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

### COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

### EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

### REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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## GETTING ORIENTED

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| 2 | Bonds and the fixed income universe    |
| 3 | Money, bond and stock markets compared |

### **B** Derivatives

|   |                                   |
|---|-----------------------------------|
| 4 | Futures: the obligation           |
| 5 | Options: the right                |
| 6 | What an option premium is made of |

### **C** Pooled vehicles

|   |                                 |
|---|---------------------------------|
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| 8 | Exchange-traded and hedge funds |
| 9 | Insurance and pension funds     |

## CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

# Module overview



This module surveys the instruments themselves. It begins with the two traditional securities a firm issues to raise money — equity, which shares ownership, and debt, which promises repayment — then turns to derivatives, which transfer risk between parties without either owning the underlying, and closes with the collective vehicles that pool small investments into large ones.

One thread runs through all nine topics: **each instrument exists because someone needed to move a different kind of risk.** Equity shares business risk, bonds isolate credit and interest rate risk, derivatives transfer price risk, and pooled vehicles spread all of it across many holders.

## Learning objectives

By the end of this module, you will be able to:

- 1 **Distinguish** equity from debt, describe the rights and returns of each, and compare the money, bond and stock markets.
- 2 **Explain** what a derivative is, distinguish futures from options, and decompose an option premium.
- 3 **Describe** how mutual funds, ETFs, hedge funds, insurance and pension funds pool capital or risk.
- 4 **Match** an instrument to the kind of risk it is designed to move, and justify the choice.

## Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

| Topic | What it covers                         | Part            | Source   | Serves  |
|-------|----------------------------------------|-----------------|----------|---------|
| 1     | Stocks: what equity actually buys you  | A · Ownership   | L8 V1    | LO 1    |
| 2     | Bonds and the fixed income universe    | A · Lending     | L8 V2    | LO 1    |
| 3     | Money, bond and stock markets compared | A · Both        | L8 V1–V2 | LO 1, 4 |
| 4     | Futures: the obligation                | B · Derivatives | L9 V1    | LO 2    |
| 5     | Options: the right                     | B · Derivatives | L9 V2    | LO 2    |
| 6     | What an option premium is made of      | B · Derivatives | L9 V2    | LO 2    |
| 7     | Mutual funds                           | C · Pooled      | L10 V1   | LO 3    |
| 8     | Exchange-traded and hedge funds        | C · Pooled      | L10 V1   | LO 3    |
| 9     | Insurance and pension funds            | C · Pooled      | L10 V2   | LO 3, 4 |

# Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

## Equity

Owned funds contributed by founders and investors, plus retained reserves.

## Preferred stock

Hybrid equity paying a fixed dividend, ranking ahead of common but carrying no vote.

## Fixed income security

An instrument promising a predetermined stream of interest plus principal.

## Callable bond

A bond the issuer may redeem early, typically at a small premium.

## Eurobond

A bond issued in a currency different from the country of issue.

## Derivative

An instrument whose value is derived entirely from an underlying asset.

## Futures contract

A binding obligation on both parties to trade at a fixed price on a fixed future date.

## Mark-to-market

Daily computation and settlement of profits and losses on open positions.

## Option

A contract granting the right, but not the obligation, to buy or sell at a strike price.

## Intrinsic value

What exercising the option immediately would earn; never negative.

## Net asset value

A fund's per-unit value: assets minus liabilities, divided by units outstanding.

## Exchange-traded fund

A fund trading intraday on an exchange, usually tracking an index passively.

## Indemnity

The principle that insurance restores a loss rather than creating profit.

## Defined benefit

A pension where the benefit is fixed and the employer bears investment risk.

## Residual claim

The shareholder's right to whatever remains after all other claimants are paid.

## Limited liability

Loss capped at the amount invested — the offset to equity's uncapped risk.

## Debenture

An unsecured corporate bond, relying on the issuer's general creditworthiness.

## Convertible bond

A bond the holder may convert into a preset number of equity shares.

## Mortgage-backed security

A tradable security created by pooling thousands of home loans.

## Contingent claim

Another name for a derivative — the payoff is contingent on the underlying.

## Margin

A good-faith deposit of roughly 3 to 6% of contract value to open a futures position.

## Clearing house

The entity standing between every buyer and seller, guaranteeing payment on default.

## Premium

The upfront price of an option, and the buyer's maximum possible loss.

## Time value

Compensation for the uncertainty remaining until expiry; grows with time.

## Rupee cost averaging

Buying more units when markets fall and fewer when they rise, through fixed periodic investment.

## Two and twenty

The hedge fund fee: 2% annual management plus 20% of profits above a hurdle.

## Subrogation

The insurer's right to recover from a third party who caused the loss.

## Defined contribution

A pension where contributions are fixed and the employee bears investment risk.

## PART A • OWNERSHIP AND LENDING

## The two traditional securities

### 1 Stocks: What Equity Actually Buys You

PART A • OWNERSHIP AND LENDING • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



#### IN ONE LINE

*Buying a share buys a fraction of a company — its future profits, its future losses, its decisions — with no promise of any return at all.*

#### Why this matters

Equity is the reference point against which every other instrument this week is defined. Bonds are understood by contrast with it, derivatives derive from it, and mutual funds pool it.

#### Where a company gets money

A firm draws on two pools. **Owned funds** — equity — come from founders and investors plus reserves retained from past profits: capital that already belongs to the business. **Borrowed funds** come from bonds and bank borrowings: capital the firm must repay with interest regardless of profit.

The choice turns on time horizon, flexibility and cost. Interest is fixed and must be paid; dividends flow only when there is a surplus to share.

#### Three rights, two return channels

Ownership grants three rights. **Vote the board** — elect the directors who oversee how the company is run. **Vote the big decisions** — a say on mergers, major acquisitions and changes to the corporate charter. **Residual claim** — after creditors, employees and tax authorities are paid, whatever remains belongs to shareholders.

Returns arrive two ways. **Capital gains**: the rise in share price. Buy Infosys at ₹1,400 and sell at ₹1,700 and the gain is ₹300 per share. **Dividends**: periodic cash payments profitable companies distribute, usually quarterly or half-yearly.

#### Common and preferred

**Common stock** is what most retail investors buy: voting rights, variable dividends dependent on profits and board discretion, full participation in price appreciation, and last place in the queue for both dividends and liquidation. **Preferred stock** is a hybrid — debt-like equity. No vote, a fixed dividend much like a bond coupon, priority over common shareholders for dividends, and a higher rank in liquidation.

#### Why equity is riskier than debt

**Lower priority**: in a default, bondholders are paid first and equity holders see nothing until every senior claim is settled. **No guarantee**: dividends are not promised and can be cut without warning. **Residual claim**: equity is paid last in liquidation, and often nothing is left. **No maturity**: a share offers no scheduled return of principal. The offset is **limited liability** — loss is capped at the amount invested. Uncapped upside, capped downside.

## COMMON STOCK AND PREFERRED STOCK • THE SAME COMPANY, DIFFERENT CONTRACTS

**COMMON STOCK****VOTING RIGHTS**

Yes — board elections and major decisions

**DIVIDENDS**

Variable; depend on profit and board discretion

**PRIORITY**

Paid after preferred, last in liquidation

**CAPITAL GAINS**

Full participation — the upside engine

**PREFERRED STOCK****VOTING RIGHTS**

None

**DIVIDENDS**

Fixed, much like a bond coupon

**PRIORITY**

Ahead of common for dividends and in liquidation

**CAPITAL GAINS**

Limited — the security behaves more like debt

*Preferred stock sits **between** debt and equity. That is why it is called a hybrid, and why exam questions use it to test whether you understand both.*

**Figure 1.1** The two forms of equity. Preferred trades voting power and upside for priority and certainty.

**KEY IDEA**

A shareholder is not owed anything. Every other instrument this week involves a promise of some kind; equity involves none. That single absence explains the higher risk, the higher expected return, and the voting rights that compensate for having no contract to enforce.

**EXAM POINTER**

“Why are stocks riskier than bonds?” wants four reasons — lower priority, no guaranteed dividend, residual claim, no maturity — and then the offset, limited liability. An answer giving only “prices fluctuate” misses the structural point, which is about the order of the payment queue rather than volatility.

**COMMON PITFALL**

Describing preferred stock as “safer equity” and stopping there. State the trade: it gains dividend priority and liquidation rank, and it surrenders the vote and most of the capital-gain participation.

**QUICK RECAP • TOPIC 1**

- Firms raise capital from owned funds (equity) or borrowed funds (debt).
- Three shareholder rights: vote the board, vote major decisions, residual claim.
- Two return channels: capital gains and dividends.
- Common stock votes and participates fully; preferred pays a fixed dividend and ranks ahead.
- Equity is riskier for four structural reasons, offset only by limited liability.

## 2 Bonds and the Fixed Income Universe

PART A • OWNERSHIP AND LENDING • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



### IN ONE LINE

*A bond is a contract: fixed payments, a fixed date, and priority over every shareholder — and the universe of them splits cleanly by tenor.*

### Why this matters

Bonds are the whole of Weeks 7 and 8. What matters here is the map — which instruments exist, who issues them, and what distinguishes one from another.

### The two neighbourhoods

**Fixed income securities** promise a predetermined stream of payments — interest plus principal — calculated on a defined formula. The universe splits in two. The **money market** handles tenors of one year or less: a temporary parking place for surplus cash, highly liquid, low risk, modest yields. The **capital bond market** handles long-term debt of one to thirty years, funding new plants, expansions and infrastructure.

### Five money market instruments

| Instrument              | What it is                                                                   | Tenor                |
|-------------------------|------------------------------------------------------------------------------|----------------------|
| Treasury bills          | Issued by governments at a discount to face value                            | 91, 182 and 364 days |
| Certificates of deposit | Bank-issued time deposits; large ones trade in secondary markets             | Under one year       |
| Commercial paper        | Unsecured promissory notes from top-rated firms, backed by bank credit lines | Maximum 270 days     |
| Repurchase agreements   | Collateralised borrowing — sell today, repurchase later at a higher price    | Very short           |
| Bankers' acceptances    | Trade finance acceptances and offshore dollar deposits                       | Under one year       |

All five share the same traits: short term, highly liquid, large denomination, low default risk and low yielding.

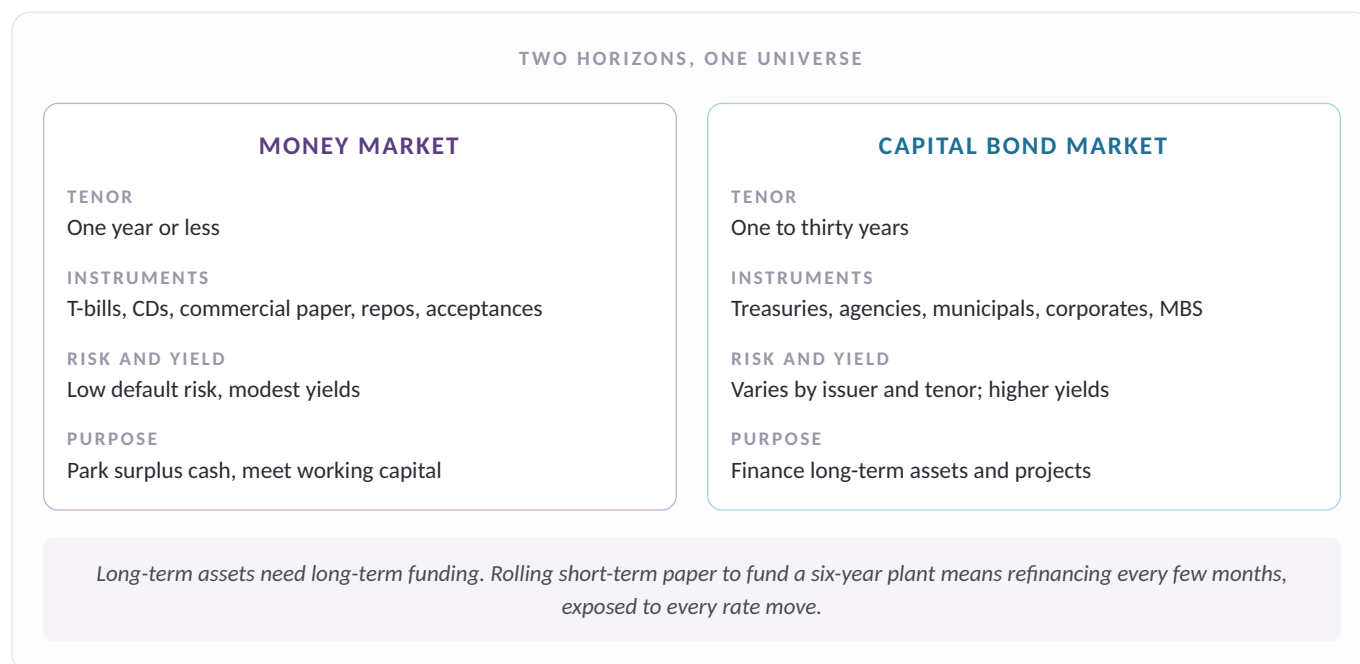
### Who issues long-term debt

**Government bonds** are the safest, backed by the sovereign. Treasury notes run to ten years, treasury bonds ten to thirty, both paying semi-annual coupons. **Agency debt** sits a notch above in yield — quasi-sovereign paper, with NABARD and HUDCO the Indian equivalents. **Municipal bonds** are state and local government debt, tax-advantaged, either general obligation (backed by taxing power) or revenue (backed by a specific project). **Corporate bonds** yield more because investors are paid for credit risk; they may be secured, unsecured debentures, subordinated, callable or convertible. **Mortgage-backed securities** pool thousands of home loans into one tradable instrument.



## International names

**Eurobonds** are issued in a currency different from the country of issue — a dollar bond sold in London. **Yankee bonds** are dollar bonds issued by foreign firms inside the United States. **Samurai bonds** are yen-denominated bonds issued by non-Japanese entities. The motive is always the same: tap whichever investor base offers the cheapest funding.



**Figure 2.1** The fixed income universe divides by tenor before anything else.

### REAL-WORLD EXAMPLE

Consider funding a six-year refinery build. Commercial paper is cheaper, but it must be rolled every few months and re-prices with every rate move. A long-term bond fixes the interest cost for the life of the project. It costs more — and it buys the planning certainty a long-lived asset demands.

### EXAM POINTER

Name all five money market instruments with tenor, and at least four categories of long-term debt with what distinguishes each. For corporate bonds, the examinable features are the five modifiers: secured, debenture, subordinated, callable, convertible — know what each changes about the contract.

### QUICK RECAP • TOPIC 2

- Fixed income promises a predetermined stream of interest plus principal.
- Money market: one year or less, liquid, low risk, low yield. Capital market: one to thirty years.
- Five money market instruments: T-bills, CDs, commercial paper, repos, bankers' acceptances.
- Long-term issuers: government, agency, municipal, corporate, mortgage-backed.
- Eurobonds, Yankee and Samurai bonds are named by where and in what currency they are issued.

## 3 Money, Bond and Stock Markets Compared

PART A • OWNERSHIP AND LENDING • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 1



### IN ONE LINE

*Three markets, one ladder: moving from money to bonds to stocks trades certainty for opportunity.*

### Why this matters

This is the consolidating topic of Part A and the comparison examiners reach for most often, because it tests whether the previous two topics were understood as a system rather than as two lists.

### The ladder

The three markets differ on every dimension that matters: how long the instrument lives, what it pays, how risky it is, and where its holder stands in the queue if the issuer fails. Read the figure across rather than down — the point is the progression, not the individual cells.

### Why the liquidation order is the spine

Money market instruments are paid first, bondholders second, shareholders last. Everything else follows from that. Being paid last is why equity demands the highest expected return; being paid first is why treasury bills yield the least. Risk and return are not arbitrary properties of these instruments — they are consequences of position in a queue.

### Limited liability, the one asymmetry

Money and bond market holders have no liability beyond their investment, and neither do shareholders. But only shareholders combine that cap with unlimited upside. The most a bondholder can receive is the contracted stream; the most a shareholder can receive has no ceiling.

## THE LADDER • CERTAINTY AT ONE END, OPPORTUNITY AT THE OTHER

|                          | Money market            | Bond market                    | Stock market                 |
|--------------------------|-------------------------|--------------------------------|------------------------------|
| <b>Maturity</b>          | One year or less        | One to thirty years            | No maturity                  |
| <b>Instruments</b>       | T-bills, CDs, CP, repos | Government and corporate bonds | Common and preferred shares  |
| <b>Purpose</b>           | Park funds short term   | Long-term financing            | Ownership and growth         |
| <b>Return type</b>       | Interest income         | Coupon payments                | Dividends plus capital gains |
| <b>Risk level</b>        | Low                     | Moderate                       | Higher and uncapped          |
| <b>Liquidation order</b> | Paid first              | Paid second                    | Paid last                    |
| <b>Liability</b>         | None                    | None                           | Limited to the stake         |

*Moving down the ladder trades **certainty for opportunity**, and stability for long-term growth.*

**Figure 3.1** The three markets side by side. The liquidation row explains every row above it.

**KEY IDEA**

Risk and return in these three markets are not independent characteristics that happen to correlate. They are both consequences of one thing: where the holder stands in the payment queue.

**EXAM POINTER**

Comparison questions want at least five axes. Maturity, instruments, return type, risk and liquidation order will always score; adding purpose and liability makes it complete. Close with the ladder sentence — certainty traded for opportunity — which shows you have read the table as a system.

**QUICK RECAP • TOPIC 3**

- Three markets differ by maturity, instruments, purpose, return type, risk and liquidation order.
- Money market instruments are paid first, bonds second, equity last.
- That queue position is what generates the risk and return differences.
- Only equity combines limited liability with an uncapped upside.
- Moving from money to bonds to stocks trades certainty for opportunity.

## PART B • DERIVATIVES

## Instruments that borrow their value

## 4 Futures: The Obligation

PART B • DERIVATIVES • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



## IN ONE LINE

*A futures contract binds both parties. There is no premium, no right to walk away, and no floor under the loss.*

### Why this matters

Derivatives are Weeks 9 and 10 in full. This topic establishes the one distinction everything else rests on: a futures contract is a promise, an option is a right.

### What a derivative is

A derivative's value is derived entirely from another asset, the *underlying* — which is why economists call them **contingent claims**. Underlyings include shares such as Infosys or Reliance, government and corporate bonds, commodities such as wheat and gold, indices such as the Nifty 50, interest rates and currencies. Four instruments dominate: futures, options, forwards and swaps.

A derivative has no intrinsic value. Its worth is borrowed entirely from the asset it points to.

### Three purposes

**Hedging** is insurance for businesses — a farmer sells futures to lock in his price, a refiner locks in crude before delivery. The motive is not profit but certainty, paid for in forfeited upside. **Speculation** is profit from a view: higher risk, higher reward, and crucially speculators supply the liquidity hedgers need. **Price discovery** pulls new information into prices faster than spot markets, and at lower cost.

### The contract itself

A futures contract is a legal requirement to buy or sell an asset at a fixed price on a specified future date. **Both parties are bound** — the buyer must buy, the seller must sell, with no opt-out. That single word, obligation, is what separates futures from options.

Contracts are **standardised**: quantity, quality and expiry are fixed by the exchange. To leave a position before expiry, a trader enters a reversing trade rather than taking delivery.

### How the exchange guarantees it

**Margin** is a good-faith deposit of roughly 3 to 6% of contract value — you never pay the full notional. **Mark-to-market** settles profits and losses at each day's close, so losses cannot pile up unnoticed. The **clearing house** steps between every buyer and seller, so you effectively trade with the exchange and payment is guaranteed even if the other side defaults. Together, counterparty default risk vanishes.

## TWO HEDGERS, OPPOSITE POSITIONS, THE SAME GOAL

**SHORT HEDGE • THE WHEAT FARMER****EXPOSURE**

Crop at harvest, facing the risk of a price fall

**ACTION**

Sells wheat futures at a known forward price

**OUTCOME**

Locks in revenue regardless of spot moves

**TRADE-OFF**

Forfeits the upside if wheat rallies past the lock

**LONG HEDGE • THE AIRLINE****EXPOSURE**

Needs fuel for months, facing the risk of a price rise

**ACTION**

Buys jet fuel futures to fix the input cost

**OUTCOME**

Locks in fuel cost regardless of spot moves

**TRADE-OFF**

Loses the benefit if fuel prices collapse later

*The farmer shorts to protect **revenue**. The airline goes long to protect **costs**. Both trade upside for predictability.*

**Figure 4.1** Hedging in practice. Opposite trades, identical motive.

**KEY IDEA**

Profit = (final price – futures price) × contract size. Because the contract size is fixed and large, a small move in the underlying produces a large rupee outcome — which is leverage, and why margin exists.

**COMMON PITFALL**

Describing a futures position as “buying at a discount”. Nothing is bought at the outset and no premium is paid. Two promises are exchanged, which is precisely why neither side pays the other anything on day one.

**EXAM POINTER**

Name the three purposes and give one example of each. For the contract, the marks sit on the word *obligation* and on the three protections — margin, mark-to-market, clearing house — with what each one does. Long profits when prices rise with unlimited upside; short profits when they fall, with unlimited loss exposure.

**QUICK RECAP • TOPIC 4**

- A derivative draws its value entirely from an underlying asset; it has no intrinsic value.
- Three purposes: hedging for certainty, speculation for profit, price discovery for efficiency.
- A futures contract is an obligation binding both parties — not a right.
- Margin, daily mark-to-market and the clearing house together eliminate counterparty risk.
- A short hedge protects revenue; a long hedge protects costs.

## 5 Options: The Right

PART B • DERIVATIVES • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 2



### IN ONE LINE

*An option grants a right and never an obligation, bought with a premium that is also the buyer's maximum possible loss.*

### Why this matters

The asymmetry between buyer and writer is the single most examined feature of options, and it is what makes every strategy in Week 10 possible.

### Four pieces

**The right.** A contract giving one party the right, but not the obligation, to buy or sell an underlying at a pre-agreed strike price. **The premium.** The small upfront sum paid for that right — the price of optionality and the buyer's maximum loss. **The asymmetry.** The buyer holds flexibility; the seller, called the *writer*, carries the obligation. **The exercise window.** The right can be used on or before a set date; if exercising is unprofitable, the buyer walks away and loses only the premium.

### Calls and puts

A **call** gives the right to buy — think of calling the shares towards you. Its intrinsic value is  $\max(\text{spot} - \text{strike}, 0)$ , and it profits when spot rises above the strike. A **put** gives the right to sell — putting the shares onto the other party. Its intrinsic value is  $\max(\text{strike} - \text{spot}, 0)$ , and it profits when spot falls below the strike. Both are floored at zero: intrinsic value can never be negative.

### Break-even, and the warning that goes with it

For a long call, break-even = strike + premium. For a long put, break-even = strike – premium. A price beyond the strike does not by itself mean profit — the premium must be recovered first. With a ₹270 strike bought for ₹12, the share reaching ₹275 still leaves the buyer ₹7 down.

### Why the payoffs are not symmetrical

A share price can rise without limit but cannot fall below zero. So a call buyer's gain is unlimited, while a put buyer's gain is large but bounded — at most the strike minus the premium, reached only if the share becomes worthless. Every option has a buyer and a writer, and what is profit for one is loss for the other. Options are zero-sum contracts.

## THE RIGHT, AND ITS MIRROR • SIX POSITIONS IN ONE VIEW

| Position      | Right or obligation | Profits when        | Maximum gain           | Maximum loss           |
|---------------|---------------------|---------------------|------------------------|------------------------|
| Call buyer    | Right to buy        | Price rises         | Unlimited              | The premium paid       |
| Call writer   | Obligation to sell  | Price flat or falls | The premium received   | Unlimited              |
| Put buyer     | Right to sell       | Price falls         | Large, bounded at zero | The premium paid       |
| Put writer    | Obligation to buy   | Price flat or rises | The premium received   | Large, bounded at zero |
| Futures long  | Obligation to buy   | Price rises         | Unlimited              | Unlimited              |
| Futures short | Obligation to sell  | Price falls         | Unlimited              | Unlimited              |

The three option pairs **mirror** each other. Futures positions take unlimited risk on both sides, because neither party can abandon the contract.

**Figure 5.1** Six positions compared. Read the last two rows against the first four to see what a premium buys.

**KEY IDEA**

The premium is not a fee. It is the price of a floor: it converts an unlimited downside into a known, bounded one. That is exactly what the futures buyer, paying nothing, does not get.

**EXAM POINTER**

Give both intrinsic value formulas and both break-evens. State the asymmetry explicitly — unlimited for a call buyer, bounded for a put buyer because price cannot go below zero. The warning that a price beyond the strike is not yet a profit is a favourite discriminator.

**QUICK RECAP • TOPIC 5**

- An option grants a right, never an obligation; the writer carries the obligation.
- Call intrinsic value is  $\max(\text{spot} - \text{strike}, 0)$ ; put is  $\max(\text{strike} - \text{spot}, 0)$ ; both floored at zero.
- Break-even is  $\text{strike} + \text{premium}$  for a call,  $\text{strike} - \text{premium}$  for a put.
- Call upside is unlimited; put upside is large but bounded because prices cannot fall below zero.
- Every gain to a buyer is a loss to the writer — options are zero-sum.

## 6 What an Option Premium Is Made Of

PART B • DERIVATIVES • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 2



### IN ONE LINE

*Every premium is intrinsic value plus time value — and understanding which is which explains why call and put prices run in opposite directions across strikes.*

### Why this matters

This is the topic that turns options from a story into a calculation, and it is the direct foundation of the moneyness and strategy work in Week 10.

### Intrinsic value, from the seller's side

View the transaction as the seller. With the spot at ₹255, a buyer asks you to deliver shares in a month at ₹200. You immediately face a disadvantage of ₹55 per share — selling for ₹200 what is worth ₹255 — and must be compensated for it. That built-in disadvantage is **intrinsic value**.

Hence the inverse relationship for calls: **low strike, high premium; high strike, low premium**. For puts it runs the other way, because a higher strike gives the put buyer a greater built-in advantage. Note that premiums still differ even where intrinsic value is zero — a ₹260 strike needs the share to move only ₹5 to become exercisable, while a ₹320 strike needs far more.

### Time value

A call seller fears the price rising well above the strike. Put yourself in that position: would you rather have five trading days of exposure remaining, or eighty? The longer the time, the greater the chance the price moves against you — so a longer period until expiry creates greater risk for the seller, and the greater the risk, the higher the premium demanded.

A put seller's fear is the mirror image: that the price falls far below the strike. Same logic, opposite direction.

### Trading the option rather than exercising it

An option contract can be bought and sold at any time before expiry, exactly like a share. So a trader need never touch the underlying to make money. Buy a ₹270 strike call at ₹2.75 with a lot size of 1,000 — an outlay of ₹2,750. If the share then rises to ₹300, the option gains substantial intrinsic value and its premium may reach roughly ₹35. Selling the contract at that point yields ₹32.25 per share, or ₹32,250, without a single share changing hands. In the Indian market most option positions are closed this way.



$$\text{PREMIUM} = \text{INTRINSIC VALUE} + \text{TIME VALUE}$$

| Contract (spot ₹255) | Market premium | Intrinsic value | Time value |
|----------------------|----------------|-----------------|------------|
| ₹240 call            | ₹18.70         | ₹15.00          | ₹3.70      |
| ₹270 call            | ₹2.75          | Nil             | ₹2.75      |
| ₹240 put             | ₹2.30          | Nil             | ₹2.30      |
| ₹270 put             | ₹18.40         | ₹15.00          | ₹3.40      |

Read down the calls, then down the puts. As the strike rises, **call premiums fall and put premiums rise** — mirrored behaviour from the same spot price.

**Figure 6.1** Four contracts on one underlying. Given any two of the three quantities, the third follows by subtraction.

#### REAL-WORLD EXAMPLE

Buy a ₹240 strike put at ₹2.30 — ₹2,300 for the lot. A week later bad news drops the share from ₹255 to ₹220. Intrinsic value alone is now ₹20, and with time still remaining the premium reaches about ₹24. Selling the contract returns ₹21.70 per share, or ₹21,700, on an initial ₹2,300.

#### EXAM POINTER

Time value = premium – intrinsic value. Compute it, do not describe it. Then state the directional rule explicitly: call premiums fall as strikes rise, put premiums rise, and both increase with time to expiry. The reason — seller's risk — carries the explanation marks.

#### QUICK RECAP · TOPIC 6

- Option premium = intrinsic value + time value.
- Call intrinsic value is  $\max(\text{spot} - \text{strike}, 0)$ ; put is  $\max(\text{strike} - \text{spot}, 0)$ .
- Call premiums fall as strikes rise; put premiums rise — mirrored behaviour.
- Time value grows with time to expiry, because the seller's risk grows with it.
- Most positions are closed by selling the contract before expiry, not by exercising.

## PART C • POOLED VEHICLES

## Many small wallets, one large portfolio

### 7 Mutual Funds

PART C • POOLED VEHICLES • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 3



#### IN ONE LINE

*A mutual fund turns many small wallets into one large, diversified, professionally run portfolio.*

#### Why this matters

Pooling is the last of the three mechanisms this week — after owning and lending directly, and after transferring risk through derivatives. It is also how most Indian retail investors actually participate.

#### Four ideas that define the vehicle

**Pooling:** money from thousands of small investors is collected into one corpus and deployed across stocks, bonds and money market instruments. **Units issued:** each investor receives units proportional to their contribution — a fractional claim on the entire portfolio. **Professional management:** trained managers handle research, selection and rebalancing. **Scale lowers cost:** pooled buying drives brokerage and transaction costs sharply down.

#### Five benefits, one structure choice

Mutual funds bridge five gaps at once: **liquidity** (units redeem quickly even when the fund holds long-term assets), **denomination** (entry from ₹500 a month), **diversification** (one unit gives exposure to dozens of securities), **cost**, and **expertise**.

Structurally they come in two forms. **Open-end** funds — roughly 98% of assets — continuously issue and redeem units on demand at end-of-day net asset value, and are not exchange listed. **Closed-end** funds issue a set number of shares at launch, trade on exchanges like ordinary equity, and their market price can drift above or below NAV.

**NAV = (total assets – total liabilities) ÷ units outstanding**, computed every business day.

#### How Indians actually invest

Through **systematic investment plans**. A fixed amount, often as little as ₹500 a month, is debited automatically into a chosen scheme. The mechanism produces *rupee cost averaging* — more units are bought when markets fall and fewer when they rise, making volatility an ally. Automation also removes the need to time markets, turning saving into a default behaviour rather than a deliberate monthly choice.

## FIVE CATEGORIES • PICK BY PURPOSE, NOT BY PAST RETURN

| Category            | Invests in                          | Objective                        | Risk     |
|---------------------|-------------------------------------|----------------------------------|----------|
| <b>Equity</b>       | Stocks and shares                   | Long-term capital appreciation   | High     |
| <b>Bond</b>         | Government and corporate bonds      | Regular income, stable returns   | Moderate |
| <b>Hybrid</b>       | A mix of stocks and bonds           | Balanced growth and income       | Moderate |
| <b>Money market</b> | Short-term money market instruments | High liquidity and safety        | Low      |
| <b>Index</b>        | The securities in a market index    | Passive, low-cost index matching | Moderate |

*The right category follows from the investor's **horizon, income need and tolerance for volatility** — not from which fund performed best last year.*

**Figure 7.1** The five fund categories and what each is for.

**KEY IDEA**

A mutual fund does not make an investor better at picking securities. It removes the need to — by supplying diversification, professional selection and institutional pricing at a scale no individual could reach alone.

**EXAM POINTER**

State the NAV formula and note it is computed daily. For open versus closed-end, the discriminating detail is that closed-end shares can trade at a premium or discount to NAV, because their price is set by the market rather than by the fund. Name all five categories with risk level.

**QUICK RECAP • TOPIC 7**

- A mutual fund pools many small investments into one diversified, professionally managed portfolio.
- Five benefits: liquidity, denomination, diversification, cost, expertise.
- Open-end funds dominate at roughly 98% of assets and price daily at NAV; closed-end funds trade on exchanges.
- $\text{NAV} = (\text{total assets} - \text{total liabilities}) \div \text{units outstanding}$ .
- SIPs produce rupee cost averaging and remove the need to time markets.

## 8 Exchange-Traded and Hedge Funds

PART C • POOLED VEHICLES • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 3



### IN ONE LINE

*An ETF is a fund that trades like a share; a hedge fund is a fund that plays by different rules and is closed to most people.*

### Why this matters

The two structural cousins of the mutual fund sit at opposite ends of the access spectrum — one designed for the widest possible participation, the other deliberately restricted.

### Exchange-traded funds

An ETF holds a portfolio like a mutual fund but trades on an exchange throughout the day, like a share of Reliance. Most replicate an index — the Nifty 50, the Sensex, or a sector index — and that passive structure means much lower annual costs than actively managed funds. Indian examples include Nippon India Nifty BeES and the SBI Nifty 50 ETF, both on the NSE.

### Hedge funds: the 2 and 20 world

**Access** is restricted to high net worth and institutional investors, with minimum ticket sizes running into crores rather than lakhs. **Strategy freedom** is the point: hedge funds may use leverage, short selling, derivatives and unlisted assets — strategies a mutual fund cannot run. The **2 and 20 fee** is a 2% annual management charge plus a 20% share of profits above an agreed hurdle rate. **Lighter regulation** brings lighter disclosure, limited liquidity windows and longer lock-ins.

They carry concentrated, leveraged risk with limited transparency. They are not a retail product.

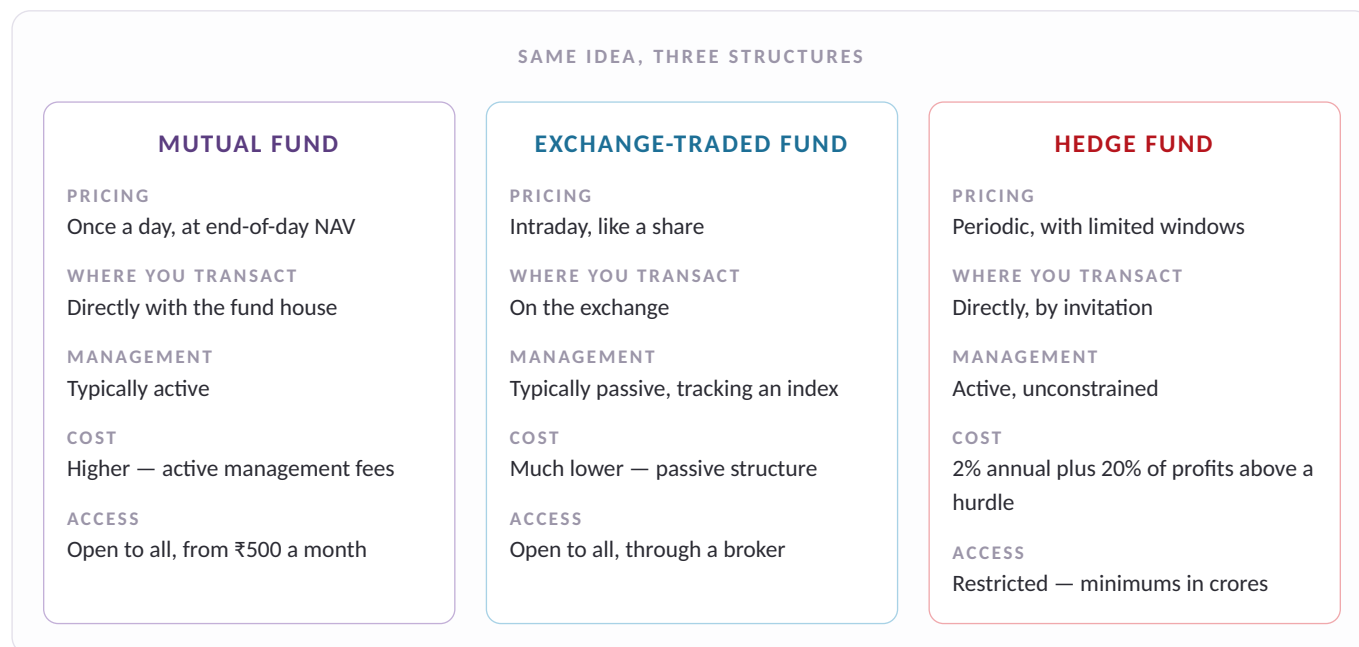


Figure 8.1 Three pooled vehicles. The differences are pricing, cost and who is allowed in.

### COMMON PITFALL

Assuming a hedge fund is simply a more sophisticated mutual fund. The defining differences are legal and structural — who may invest, what strategies are permitted, and how much must be disclosed — not the skill of the manager.

**EXAM POINTER**

The ETF comparison wants pricing (intraday versus end-of-day NAV), management style (passive versus active) and cost, with one Indian example. For hedge funds, name the 2 and 20 structure and at least two of the four access or strategy features.

**QUICK RECAP • TOPIC 8**

- An ETF trades intraday on an exchange and usually tracks an index passively at low cost.
- Indian examples include Nippon India Nifty BeES and the SBI Nifty 50 ETF.
- Hedge funds are restricted to high net worth and institutional investors.
- They may use leverage, short selling, derivatives and unlisted assets.
- The 2 and 20 fee is 2% annual management plus 20% of profits above a hurdle rate.

## 9 Insurance and Pension Funds

PART C • POOLED VEHICLES • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 3



### IN ONE LINE

*Insurance pools risk to avoid a loss; a pension pools savings to fund a retirement. Both turn many uncertain individual outcomes into one predictable group outcome.*

### Why this matters

The closing topic of the week, and the one that distinguishes pooling for *protection* from pooling for *accumulation* — a distinction examiners test precisely because the mechanisms look so similar.

### Why insurance exists

**Risk aversion.** Given a choice between a small certain payment and a huge uncertain loss, most people prefer the small certain payment. That preference is the foundation of every policy. **The business model** is that insurers assume risk in exchange for premiums, profiting when premiums collected exceed claims and expenses paid. **Risk pooling** makes it work: across ten lakh policyholders the average loss becomes remarkably predictable, turning many uncertain individual risks into one calculable group outcome.

### The six principles

**Insurable interest:** a genuine financial link must exist between the insured and the beneficiary — your own life, not a stranger's. **Utmost good faith:** full disclosure is required, and hiding a material fact can void the policy. **Indemnity:** insurance restores a loss, it does not create profit. **Subrogation:** recovery from a third party who caused the loss reduces the insurer's liability. **Risk pooling:** risk must spread across many policyholders. **Calculability:** losses must be measurable and probabilities predictable, or no actuary can price a premium.

Together these six separate insurance from gambling. Each protects the integrity of the pool.

### Pension funds, and the shift in who carries the risk

A pension plan is an asset pool built during working years and paid out as income in retirement. India's twin pillars are the **National Pension System**, a defined contribution scheme open to all citizens and portable across jobs, and the **Employees' Provident Fund Organisation**, administering mandatory savings for the formal sector.

Four forces drive growth: *urbanisation* weakens the rural model of children supporting elderly parents; *longer lives* stretch retirement across two or three decades; *earlier retirement* compresses the earning years; and the *shift away from the joint family* means retirement security must be built individually.

## TWO DESIGNS, AND THE RISK THAT MOVED

**DEFINED BENEFIT****RETIREMENT BENEFIT**

Fixed or guaranteed

**CONTRIBUTION**

The employer adjusts as needed

**INVESTMENT RISK**

Borne by the employer

**BENEFIT BASIS**

Salary plus years of service

**MAIN RISK**

Plan underfunding

**TREND**

Declining globally

**DEFINED CONTRIBUTION****RETIREMENT BENEFIT**

Depends on investment performance

**CONTRIBUTION**

Fixed contributions are specified

**INVESTMENT RISK**

Borne by the employee

**BENEFIT BASIS**

Account balance at retirement

**MAIN RISK**

Poor investment decisions

**TREND**

Increasing — India's NPS is one

*Investment risk has moved from employer to employee. That single shift is why defined contribution is now the rising model worldwide.*

**Figure 9.1** The two pension designs. The row that matters is who carries the investment risk.

**KEY IDEA**

Insurance and mutual funds both pool, but for opposite purposes. Insurance pools to **avoid a loss**; a mutual fund pools to **accumulate wealth**. In insurance the objective is not profit but the reduction of a loss when a risk event materialises.

**EXAM POINTER**

Name all six insurance principles — they are frequently asked as a complete list. For pensions, the defined benefit versus defined contribution comparison should be answered on who bears the investment risk, what determines the benefit, and the direction of the global trend.

**QUICK RECAP • TOPIC 9**

- Insurance exists because people prefer a small certain payment to a large uncertain loss.
- Risk pooling turns many uncertain individual risks into one calculable group outcome.
- Six principles: insurable interest, utmost good faith, indemnity, subrogation, risk pooling, calculability.
- Insurance pools for loss avoidance; mutual funds pool for wealth accumulation.
- Defined benefit puts investment risk on the employer; defined contribution puts it on the employee.

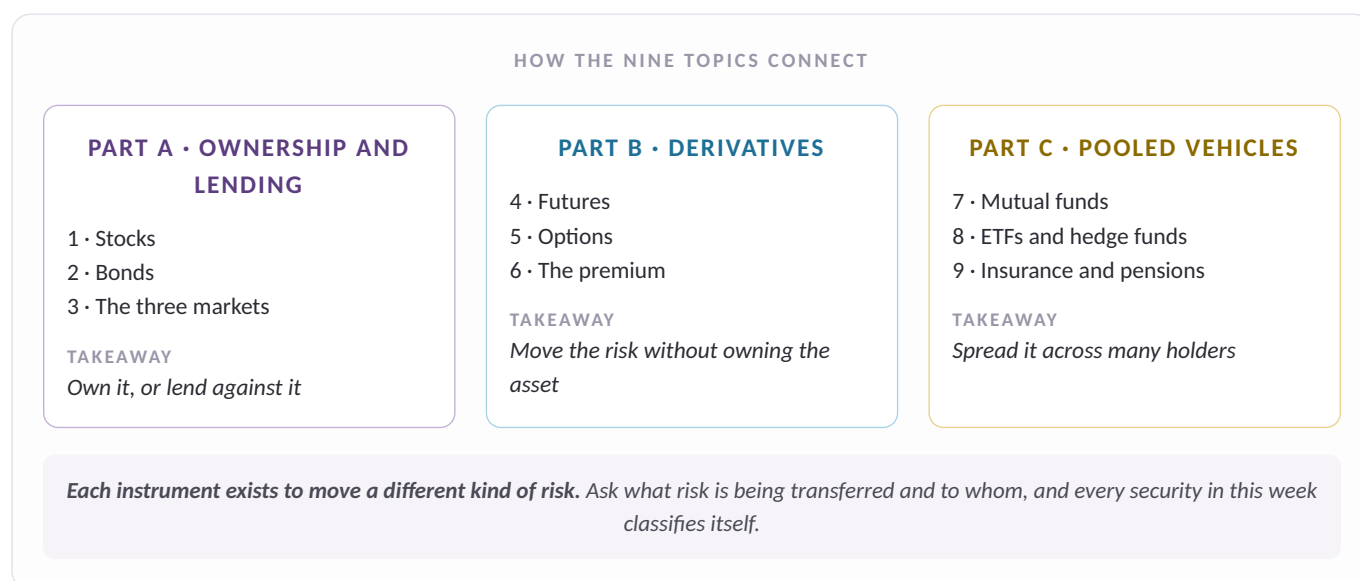
## Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

| LO | What you should now be able to do                                          | Where it was covered                      |
|----|----------------------------------------------------------------------------|-------------------------------------------|
| 1  | Distinguish equity from debt and compare the three markets                 | Topics 1, 2 and 3 · Figures 1.1, 2.1, 3.1 |
| 2  | Explain derivatives, distinguish futures from options, decompose a premium | Topics 4, 5 and 6 · Figures 4.1, 5.1, 6.1 |
| 3  | Describe how pooled vehicles gather capital or risk                        | Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1 |
| 4  | Match an instrument to the risk it is designed to move                     | Topics 3 and 9 · Figures 3.1, 9.1         |

### The module in six sentences

Equity buys ownership — three rights, two return channels, and no promise of anything — while debt buys a contract with fixed payments and priority in the queue. The fixed income universe divides first by tenor, with the money market handling a year or less and the capital market funding assets that live for decades. Moving from money to bonds to stocks trades certainty for opportunity, and the liquidation order explains why. Derivatives borrow their value entirely from an underlying, and split on one word: a futures contract obliges both parties, while an option grants a right that can be abandoned for the cost of the premium. Every premium is intrinsic value plus time value, which is why call and put prices run in opposite directions across strikes. Pooled vehicles gather what individuals cannot — mutual funds and ETFs pool capital to accumulate wealth, while insurance and pension funds pool risk and savings to protect against a loss and fund a retirement.



**Figure M1** How the nine topics connect. Revise along the parts, not down the list.



# Glossary and self-check



| Term                            | Definition                                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Equity</b>                   | Owned funds contributed by founders and investors, plus retained reserves.                       |
| <b>Residual claim</b>           | The shareholder's right to whatever remains after all other claimants are paid.                  |
| <b>Preferred stock</b>          | Hybrid equity paying a fixed dividend, ranking ahead of common but carrying no vote.             |
| <b>Limited liability</b>        | Loss capped at the amount invested — the offset to equity's uncapped risk.                       |
| <b>Fixed income security</b>    | An instrument promising a predetermined stream of interest plus principal.                       |
| <b>Debenture</b>                | An unsecured corporate bond, relying on the issuer's general creditworthiness.                   |
| <b>Callable bond</b>            | A bond the issuer may redeem early, typically at a small premium.                                |
| <b>Convertible bond</b>         | A bond the holder may convert into a preset number of equity shares.                             |
| <b>Eurobond</b>                 | A bond issued in a currency different from the country of issue.                                 |
| <b>Mortgage-backed security</b> | A tradable security created by pooling thousands of home loans.                                  |
| <b>Derivative</b>               | An instrument whose value is derived entirely from an underlying asset.                          |
| <b>Contingent claim</b>         | Another name for a derivative — the payoff is contingent on the underlying.                      |
| <b>Futures contract</b>         | A binding obligation on both parties to trade at a fixed price on a fixed future date.           |
| <b>Margin</b>                   | A good-faith deposit of roughly 3 to 6% of contract value to open a futures position.            |
| <b>Mark-to-market</b>           | Daily computation and settlement of profits and losses on open positions.                        |
| <b>Clearing house</b>           | The entity standing between every buyer and seller, guaranteeing payment on default.             |
| <b>Option</b>                   | A contract granting the right, but not the obligation, to buy or sell at a strike price.         |
| <b>Premium</b>                  | The upfront price of an option, and the buyer's maximum possible loss.                           |
| <b>Intrinsic value</b>          | What exercising the option immediately would earn; never negative.                               |
| <b>Time value</b>               | Compensation for the uncertainty remaining until expiry; grows with time.                        |
| <b>Net asset value</b>          | A fund's per-unit value: assets minus liabilities, divided by units outstanding.                 |
| <b>Rupee cost averaging</b>     | Buying more units when markets fall and fewer when they rise, through fixed periodic investment. |
| <b>Exchange-traded fund</b>     | A fund trading intraday on an exchange, usually tracking an index passively.                     |

| Term                        | Definition                                                                      |
|-----------------------------|---------------------------------------------------------------------------------|
| <b>Two and twenty</b>       | The hedge fund fee: 2% annual management plus 20% of profits above a hurdle.    |
| <b>Indemnity</b>            | The principle that insurance restores a loss rather than creating profit.       |
| <b>Subrogation</b>          | The insurer's right to recover from a third party who caused the loss.          |
| <b>Defined benefit</b>      | A pension where the benefit is fixed and the employer bears investment risk.    |
| <b>Defined contribution</b> | A pension where contributions are fixed and the employee bears investment risk. |

## Self-check

Ungraded retrieval practice. Answer from memory first, then check.

**Q1** A company is wound up. In what order are the money market lender, the bondholder, the preferred shareholder and the common shareholder paid?

**Money market lender, bondholder, preferred shareholder, common shareholder.** That queue is the reason equity demands the highest expected return — risk and return here are consequences of position, not independent properties.

**Q2** A call option has a strike of ₹270 and a premium of ₹12. The share closes at ₹275 on expiry. Has the buyer profited?

**No.** Intrinsic value is ₹5, but ₹12 was paid, leaving a net loss of ₹7 per share. Break-even is strike + premium = ₹282. A price beyond the strike is not yet a profit.

**Q3** The spot price is ₹255. A ₹270 put trades at ₹18.40. What is its time value?

**₹3.40.** Intrinsic value =  $\max(270 - 255, 0) = ₹15$ . Time value = premium - intrinsic value =  $18.40 - 15.00$ . Note that the equivalent ₹270 call has no intrinsic value at all — the mirror behaviour.

**Q4** Why is a put buyer's maximum gain bounded when a call buyer's is not?

**Because a share price cannot fall below zero.** The most a put can be worth is the strike itself, less the premium paid. A share price can rise without any ceiling, so the call buyer's gain has none either.

**Q5** Both a mutual fund and an insurance policy pool money from many people. What is the difference in purpose?

**Accumulation versus protection.** A mutual fund pools capital to build wealth. Insurance pools risk to reduce a loss when an adverse event occurs — the objective is not profit at all.

**Q6** An employer moves its staff from a defined benefit to a defined contribution pension. Who is affected, and how?

**The employee now carries the investment risk.** Under defined benefit the payout is fixed and the employer bears any shortfall. Under defined contribution the contribution is fixed and the eventual benefit depends on investment performance — which is why the model is spreading globally.